



CyberPatriot Windows 11 Exhibition Round Image Answer Key



Welcome to the CyberPatriot Exhibition Round!

The vulnerabilities in this image are some of the most basic ones found during a CyberPatriot competition. Even if you do very well with these vulnerabilities, you will experience greater difficulty as the season progresses. The README file on the desktop in this image may be more detailed than those you see during the competition. You will have to use your own knowledge, not just the hints in this file, to achieve a high score during the actual competition.

Below are the answers to the problems that are being scored in this image.

It is also possible to lose points during the competition. Simple penalties that may arise are noted below the answers. There are many ways to solve some of the problems below. This answer key just shows one method in each case.

Coaches will be sent categories of vulnerabilities following each online round.

Answers

1. Forensics Question 1 Correct: 7 pts

- How do I find this problem?

When you open an image, please read all the "Forensics Questions" thoroughly before modifying the image as you may change something that prevents you from answering the question correctly. There is a file on the Desktop here named "Forensics Question 1".

- How do I solve this problem?

Open **attack.pcap** shortcut on Desktop > go to lines 5, 6, and 9 > host IP address is under the **Source** column

Answer: 192.168.202.138

- Why is fixing this problem important?

It is important to find out which host was exploited.

2. Forensics Question 2 Correct: 7 pts

- How do I find this problem?

When you open an image, please read all the "Forensics Questions" thoroughly before modifying the image as you may change something that prevents you from answering the question correctly. There is a file on the Desktop here named "Forensics Question 2".

- How do I solve this problem?

Open a ciphertext decryption tool such as

https://www.mobilefish.com/services/one_time_pad/one_time_pad.php > set conversion method to **Decrypt** > enter e302a1b9e317659052c83f56033d4a85c74b12e85ef63b628b82fb02b10ab1 as the **one-time pad ciphertext** > enter b76ac4cb863704e237e85e3a745c33f6e73f7b8b359f5505abe1976dd261c2 as the **message/cipher** > select **Convert**

Answer: There are always ticking clocks

- Why is fixing this problem important?

Decrypting messages using the one-time pad helps you understand the basic rules of encryption which is necessary to send secure messages across networks.

3. The Guest account is not enabled: 5 pts

- How do I find this problem?

In the search bar, search Computer Management > Local Users and Groups > Users. The Guest account is located there.

- How do I solve this problem?

In the search bar, search Computer Management > Local Users and Groups > Users > double-click on Guest > Select Account is disabled > Click Apply

- Why is fixing this problem important?

It is important because the guest account can be an entry point into a system, so disabling it is good security practice.

4. Remove unauthorized user accounts (darkarmy and whiterose): 6 pts

- How do I find this problem?
One of the first things you should do when starting an image during a competition is check the README file on the desktop. There, you will see the authorized users for the image. These are the only users that should have an account. All others should be removed.
- How do I solve this problem?
In the search box, type and select Control Panel. Select User Accounts → select Manage another account. In this window, you can click the users that are not listed on the authorized users list in the README file and select the option to “Delete the account.” Make sure to write down the names of any user you deleted. You may need this information later. You will then be prompted to delete or keep this user’s files before you delete the account. Select Delete Files → Delete Account.
- Why is fixing this problem important?
Computer access should be limited to only those who need to use it to complete their tasks. By leaving these user accounts on the image, invalid individuals may be able to log on to the computer and make changes that could affect the safety and security of legitimate users.

5. Create user account for user penguru

- How do I find this problem?
The README requests that you create a user account for penguru.
- How do I solve this problem?
In the search bar, search **Computer Management > Local Users and Groups > Users > More Actions > New User >** in the User name: field, enter **penguru > Create**
- Why is fixing this problem important?
It is important because as an admin you may need to know how to do administrative tasks when requested. The knowledge of how to do these tasks is also important because you must know how operating systems work before you can defend them.

7. Change unauthorized administrators to standard users: 8 pts

- How do I find this problem?

One of the first things you should do when starting an image during a competition is check the README file on the Desktop. The README contains authorized users for the image and the account type for each user.

- How do I solve this problem?

In the search box, type and select Control Panel. Click on User Accounts → User Accounts → Manage another account. Find the users that have an Administrator account who is listed only as a Standard user in the README file. Select Change the account type → select Standard User → select Change Account Type. Make sure to write down the names of the users you make changes to or delete. You may need this information later.

- Why is fixing this problem important?

Ensuring account types are set correctly is very important. A Standard user given administrative permissions can accidentally or purposefully cause significant damage to a system because they would have unrestricted full read and write access to all files on the system, not just their own.

8. User sswailem password expires: 5 pts

- How do I find this problem?

In the search bar, search **Computer Management > Local Users and Groups > Users** > Double-click on user **sswailem** > uncheck the box **Password never expires** > **Apply** > **OK**

- How do I solve this problem?

In the search bar, search **Computer Management > Local Users and Groups > Users** > Double-click on user **sswailem** > uncheck the box **Password never expires** > **Apply** > **OK**

- Why is fixing this problem important?

It is important because as an admin you may need to know how to do administrative tasks when requested. The knowledge of how to do these tasks is also important because you must know how operating systems work before you can defend them.

10. A secure minimum password length is required: 4 pts

- How do I find this problem?

One of the first things you should do when starting an image during a competition is check the README file on the Desktop. The README contains important information related to password policies.

- How do I solve this problem?

Press the **Windows button + R** and search **secpol.msc > Account Policies > Password Policy > Minimum password length** > Use the arrow to set the length to 12 characters > **Apply**

- Why is fixing this problem important?

A secure minimum password length is important because it prevents users from selecting simple passwords that are easy to guess or figure out which can compromise the safety of the system.

11. **A secure lockout threshold exists: 4 pts**

- How do I find this problem?
Security settings in the Local Security Policy allow the administrator to set Account Policy, Audit Policy, User Rights Assignment and security settings. Account Lockout Policy appears in Account Policies.
- How do I solve this problem?
Press the **Windows key + R** → type **secpol.msc** → Press OK → Under **Security Settings**, select **Account Policies** → select **Account Lockout Policy** → select **Account lockout threshold** → in the “invalid login attempts” field, enter a number greater than 4 and less than 51. Then, press OK.
- Why is fixing this problem important?
Setting a secure lockout threshold will ensure that a brute force password attack will result in the attacker being locked out of the account. The number of invalid login attempts should be low enough to prevent an attacker from accessing the account, but high enough so that users are not locked out of the account. Giving users at least 5 attempts prevents them from accidentally being locked out of their account.

12. **Passwords must meet complexity requirements: 4 pts**

- How do I find this problem?
Security settings in the Local Security Policy allow the administrator to set Account Policy, Audit Policy, User Rights Assignment and security settings.
Password Policy appears in Account Policies.
- How do I solve this problem?
Press the **Windows button + R** and search **secpol.msc > Account Policies > Password Policy > Password must meet complexity requirements**
- Why is fixing this problem important?
Having a weak password on a user account makes it extremely vulnerable to attacks by outside individuals. With a weak password, an attacker can more easily gain access to a user’s files. Strong passwords make it much more likely that only the authorized user of the account can access it.

13. **Microsoft network client: Digitally sign communications (always) [enabled]: 5 pts**

- How do I find this problem?
Security settings in the Local Security Policy allow the administrator to set Security Options
- How do I solve this problem?
Press the **Windows button + R** and search **secpol.msc > Local Policies > Security Options > Microsoft network client: Digitally sign communications always (always) > Enabled > Apply > OK**
- Why is fixing this problem important?
It is important to enable this option because it reduces the risk of tampering and attacks on the network, protects file integrity, and hardens Windows against certain internal-network attacks.

14. Do not require CTRL+ALT+DEL [disabled]: 5 pts

- How do I find this problem?
Security settings in the Local Security Policy allow the administrator to set Security Options
- How do I solve this problem?
Press the **Windows button + R** and search **secpol.msc > Local Policies > Security Options > Do not require CTRL+ALT+DEL > Disabled > Apply**
- Why is fixing this problem important?
It is important because it shows that the user is on a real Windows screen because it activates the Secure Attention Sequence (SAS). This is important because malware cannot easily fake this sequence. This prevents users from entering sensitive information into a potentially malicious program or software.

15. Firewall protection has been enabled: 4 pts

- How do I find this problem?
Search for Firewall & Network protection in the Windows Search bar.
- How do I solve this problem?
In the search bar, search **Firewall & network protection** > Under **Domain network**, select **Turn On > Yes** > Under **Private network**, select **Turn On > Yes** > Under **Public network**, select **Turn On > Yes**
- Why is fixing this problem important?
Firewalls are your first line of defense against attacks. You can customize your firewall settings to allow traffic for specific programs. The two most common exceptions you can create are for ports or programs.

16. Remote desktop sharing is turned off: 5 pts

- How do I find this problem?
Search up settings in the windows search bar. Then look in the Systems tab and scroll down until you find **Remote Desktop**.
- How do I solve this problem?
Search up settings in the windows search bar. Then look in the systems tab and scroll down until you find **Remote Desktop**. Select Remote Desktop and on the new page toggle the **Remote Desktop** switch to the "Off" position.
- Why is fixing this problem important?
Turning this option off is important because it decreases the amount of exposure a system has to unauthorized access and malicious attacks.

17. Notepad++ has been updated: 3 pts

- How do I find this problem?
The README states that Notepad++ is a business related software and it should be kept installed and up-to-date.
- How do I solve this problem?
Open Notepad++ > ? (Help) > **Update Notepad++** > **Yes** > **Yes** > **Yes** > **OK** > **Next** > **I Agree** > **Next** > **Next** > **Install** > **Finish**
- Why is fixing this problem important?
When security vulnerabilities are found in software, the software vendor publishes updates that patch the security vulnerabilities. Adversaries can more easily compromise your system if software is present that has known security vulnerabilities. Ensuring software is up-to-date removes known security vulnerabilities.

18. Google Chrome has been updated: 3 pts

- How do I find this problem?
The ReadMe states that the latest stable version of Firefox is required by the organization's management
- How do I solve this problem?
Open Google Chrome > On the top-right, select **Reinstall Chrome** > **Download Chrome** > On the bottom task bar of the browser, the file **ChromeSetup.exe** will appear. Select the up arrow next to the file. Then select Show in folder. Double-click on **ChromeSetup** > **Yes** > Open Google Chrome > **About Chrome** > **Relaunch**
- Why is fixing this problem important?
Browser updates often contain security features to keep up to date with new methods of attacks and repair vulnerabilities found in previous versions. Updates are also generally better at detecting malware, trojan viruses, and other types of malware, which makes your computer less susceptible to these types of attacks.

19. Removed shellshock exploit script: 5 pts

- How do I find this problem?
The README specifically states that non-work related "hacking tools" are prohibited. There are several ways and commands that can be used to find files and file types.
- How do I solve this problem?
Open File Explorer > **Local Disk (C:)** > **Users** > **ealderson** > **Downloads** > Right-click on **shellshock-exploit.py** > **Delete**
- Why is fixing this problem important?
Keeping non-work related files or hacking tools on the computer is a violation of the company's policies as mentioned in the README file.

20. Remove Jellyfin Media Player: 4 pts

- How do I find this problem?
The README file states that non-work related software is prohibited on this image.
- How do I solve this problem?
Open File Explorer > **Local Disk (C:)** > **Program Files** > Right-click **Jellyfin** folder > **Delete**
- Why is fixing this problem important?
Keeping non-work related files or hacking tools on the computer is a violation of the company's policies as mentioned in the README file.

21. Created group allsafe: 3 pts

- How do I find this problem?
The README states to create a group called allsafe
- How do I solve this problem?
Press the **Windows button + R** and search **lusrmgr.msc** > **Groups** > right click and select **New Group** > enter group name > Click Create
- Why is fixing this problem important?
It is important because as an admin you may need to know how to do administrative tasks when requested. The knowledge of how to do these tasks is also important because you must know how operating systems work before you can defend them.

22. Added users to group allsafe (ggoddard, ealderson, amoss, lchong): 5 pts

- How do I find this problem?
The README states to add the users ggoddard, ealderson, amoss, and lchong to the allsafe group.
- How do I solve this problem?
Press the **Windows button + R** and search **lusrmgr.msc** > **Groups** > right click on the **allsafe** group > select Add to Group > Click the Add... button and type each name in > Click Apply and then OK
- Why is fixing this problem important?
It is important because as an admin you may need to know how to do administrative tasks when requested. The knowledge of how to do these tasks is also important because you must know how operating systems work before you can defend them.

23. Remove Python3: 4 pts

- How do I find this problem?
The README specifically states that non-work related “hacking tools” are prohibited. There are several ways and commands that can be used to find files and file types.
- How do I solve this problem?
Go to the windows search bar and search “Add or remove programs” > click it to open it > Look for Python3 in the Installed Apps list > Click the 3 dots on the right and select **Uninstall**
- Why is fixing this problem important?
Keeping non-work related files or hacking tools on the computer is a violation of the company's policies as mentioned in the README file.

Penalties

- Account lockout threshold less than 5 is deprecated
- Google Chrome is not installed at the default location
- Notepad++ is not installed at the default location
- 7-Zip is not installed at the default location
- Wireshark is not installed at the default location